A Project Work II Report on

ARTIFICIAL INTELLIGENCE APPROACHES FOR DETECTING INHERITED RETINAL DISEASES

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

IN

DEPARTMENT OF COMPUTER ENGINEERING

by

B. BRAHMANI 160622740006

JADHAV RUPALI 160622740015

M. VARSHITHA 160622740031

Under the Esteemed Guidance of

G. PAVANI

Assistant Professor

DEPARTMENT OF COMPUTER ENGINEERING

STANLEY COLLEGE OF ENGINEERING AND TECHNOLOGY FOR WOMEN

Chapel Road, Abids, Hyderabad.

2025-2026

STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN

Chapel Road, Abids Hyderabad

DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify Project that report entitled **Title of the project** submitted by **Ms. B.BRAHMANI** (160622740006)**, Ms. JADHAV RUPALI** (160622740015), **Ms. M.VARSHITHA** (160622740031), student of **DEPARTMENT OF COMPUTER ENGINEERING**, Stanley college of Engineering and Technology for Women in partial fulfillment of the requirements for the award of the degree of Bachelor's in Engineering with **COMPUTER ENGINEERING** as specialization is a record of bonafide work carried out in IV year VIII Sem during the academic year 2025-2026.

Guide
G. Pavani
Assistant Professor
Department of CME

Head of the Dept.
Dr. K. Vaidehi
Professor & Head
Department of CME

Coordinator
Y. Gnyana Deepa
Assistant Professor
Department of CME

External Examiner

DECLARATION

We hereby certify that the work which is being presented in the project entitled **"ARTIFICIAL INTELLIGENCE APPROACHES FOR DETECTING INHERITED RETINAL DISEASES"** submitted in the **DEPARTMENT OF COMPUTER ENGINEERING** is an authentic record of my own work carried under the supervision of **G. PAVANI (**Assistant Professor **)** This study has not been submitted to any other institution or university for the award of any other degree.

B. BRAHMANI	**160622740006**
JADHAV RUPALI	**160622740015**
M. VARSHITHA	**160622740031**

ACKNOWLEDGEMENT

We are grateful to The Almighty God for establishing us to complete this project. We pay our respect and love to our parents and all our family members and friends for their love and encouragement throughout our career.

We wish to extend my deepest gratitude to all those who have supported me throughout this Project contributed to the successful completion of my Project report.

We wish to express our sincere thanks to **Sri. Kodali Krishna Rao, Correspondent and Secretary**, Stanley College of Engineering & Technology for Women, for providing us with all the necessary facilities.

We place on record our sincere gratitude **to Prof. B. L. Raju, Principal,** for his constant encouragement.

We deeply express our sincere thanks to our **Head of the Department, Prof. Dr. K.Vaidehi**, for encouraging and allowing us to present the internship on the topic "**ARTIFICIAL INTELLIGENCE APPROACHES FOR DETECTING INHERITED RETINAL DISEASES**" at our department premises for the partial fulfilment of the requirements leading to the award of the B.E. degree.

It is our privilege to express sincere regards to **our Project guide Ms. G. Pavani and Project coordinator Ms. Y. Gnyana Deepa,** for the valuable inputs, guidance, encouragement, whole hearted co-operation and constructive criticism throughout the duration of our internship.

We also express our sincere gratitude to the project coordinator for the constant support, guidance, and encouragement provided during the entire course of our project, which greatly contributed to its successful completion.

We sincerely acknowledge them for extending their valuable guidance, support for literature, critical reviews of project and the report and above all the moral support they had provided me with all stages of this project. However, it would not have been possible without the kind of support and help of many individuals. We would like to extend my sincere thanks to all of them. We are highly indebted to our mentor for their guidance and constant super vision as well as for providing necessary information regarding the project and also for their support in completing the project.

We take this opportunity to thank all our faculty, who have directly or indirectly helped our project. Last but not least, we express our thanks to our friends for their cooperation and support.

STANLEY COLLEGE OF ENGINEERING AND TECHNOLOGY FOR WOMEN

(Approved by AICTE, Accredited by NAAC 'A', UGC Autonomous)

Abids, Hyderabad, Telangana–500001

DEPARTMENT OF COMPUTER ENGINEERING

Institute Vision :

Empower Women; Impact the World. Empowering girl students through professional education integrated with values and character to make an impact in the World.

Institute Mission :

1. Providing quality engineering education for girl students to make them competent and confident to succeed in professional practice and advanced learning. Establish state-of-art-facilities and resources to facilitate world class education.

2. Integrating qualities like humanity, social values, ethics, leadership in order to encourage contribution to society.

Department Vision:

Empowering girl students with comprehensive and pragmatic knowledge in computer engineering for success in life.

Department Mission:

1.To impart quality education for girl students to learn and practice various hardware and software and to facilitate networking platforms which are the state of the art in industry. platforms prevalent in industry.

2.To achieve self-sustainability and overall development through entrepreneurship and technology transfer through research and development activities.

3.To groom students with good an optimistic attitude, excellent team work and great personality skills.

Program Educational Objectives: (PEO's)

PEO1: Our graduates shall have enhanced skills and comprehensive knowledge in software and hardware, networking technologies for professional excellence, towards successful self- employment, advanced learning, entrepreneurship and research.

PEO2: Our graduates shall have life-long learning attitude, innovation and creativity to master the state of the art technologies with inclination towards research, devising pragmatic solutions for realistic and social issues in the society.

PEO3: Our graduates shall have optimistic attitude and vibrant personality skills, high ethical values, individuality, excellent teamwork, leadership and entrepreneurial skills towards computer professionalism and ethical practices within the organization and the society.

Program Outcomes

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

Program Specific Outcomes (PSOs):

PSO1: Problem-Solving Skills: The ability to apply industry standard practices and pragmatic strategies in software and hardware and network project development using open-ended programming environments to deliver a quality product within time and budget for the benefit of students.
PSO2: Design, Implement, Test and Evaluate a computer system, software, hardware, networks, component or innovative algorithm to meet desired needs and to solve a computational problem within time and space.

Course Outcomes: At the end of this course, the student will be able to

1. To understand project characteristics and various stages of project.
2. Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.
3. Evaluate different solutions based on economic and technical feasibility.
4. Effectively plan a project and confidently perform all aspects of project management.
5. Demonstrate effective written and oral communication skills.

COPO Mapping:

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
SPW821CM.1	2	2	1	2	2	2	3	3	2	2	3	2	2
SPW821CM.2	3	3	3	2	3	2	2	2	2	3	3	3	3
SPW821CM.3	1	2	2	2	2	1	2	2	3	2	3	2	2
SPW821CM.4	1	1	1	1	2	1	2	3	3	2	2	1	1
SPW821CM.5	2	2	3	2	3	3	3	3	2	3	3	3	3

Annexure I: Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

WK → PO Mapping

WK	Mapped POs
WK1	PO1
WK2	PO1, PO2, PO4
WK3	PO1, PO2, PO3
WK4	PO2, PO3, PO4
WK5	PO3, PO7
WK6	PO5, PO11
WK7	PO6, PO7, PO10, PO11
WK8	PO4, PO12
WK9	PO8, PO9, PO12

Justification for WK → PO Mapping

WK1 → PO1

WK1 provides the scientific foundation required for application of engineering knowledge in PO1.

WK2 → PO1, PO2, PO4

Mathematical modelling and computational tools (WK2) are essential for solving problems (PO1), analyzing systems (PO2), and conducting investigations (PO4).

WK3 → PO1, PO2, PO3

Engineering fundamentals (WK3) enable knowledge application (PO1), structured problem analysis (PO2), and systematic design of solutions (PO3).

WK4 → PO2, PO3, PO4

Specialist knowledge (WK4) supports advanced problem-solving (PO2), complex system design (PO3), and technical investigations (PO4).

WK5 → PO3, PO7

Design considerations involving sustainability and lifecycle thinking (WK5) align with solution development (PO3) and environmental responsibility (PO7).

WK6 → PO5, PO11

Engineering practice knowledge (WK6) underpins modern tool usage (PO5) and project management applications (PO11).

WK7 → PO6, PO7, PO10, PO11

Societal awareness and responsibility (WK7) relate to societal engineering roles (PO6), sustainability (PO7), professional communication (PO10), and management decisions (PO11).

WK8 → PO4, PO12

Research engagement and critical thinking (WK8) are required for investigations (PO4) and life-long learning (PO12).

WK9 → PO8, PO9, PO12

Ethics and inclusive professional conduct (WK9) align with ethical practice (PO8), team work (PO9), and professional growth (PO12)

List of all 17 United Nations Sustainable Development Goals (SDGs):

SDG No.	Goal Title
SDG 1	No Poverty
SDG 2	Zero Hunger
SDG 3	Good Health and Well-being
SDG 4	Quality Education
SDG 5	Gender Equality
SDG 6	Clean Water and Sanitation
SDG 7	Affordable and Clean Energy
SDG 8	Decent Work and Economic Growth
SDG 9	Industry, Innovation and Infrastructure
SDG 10	Reduced Inequalities
SDG 11	Sustainable Cities and Communities
SDG 12	Responsible Consumption and Production
SDG 13	Climate Action
SDG 14	Life Below Water
SDG 15	Life on Land
SDG 16	Peace, Justice and Strong Institutions
SDG 17	Partnerships for the Goals

Project Title: "ARTIFICIAL INTELLIGENCE APPROACHES FOR DETECTING INHERITED RETINAL DISEASES"

Mapped Sustainable Development Goals (SDGs):

SDG No.	SDG Title	Relevance to Project
3	Good Health and Well-being	The project promotes early detection and diagnosis of inherited retinal diseases using AI, improving clinical outcomes and reducing preventable visual disability.
4	Quality Education	Supports advanced learning in AI, medical imaging, and ophthalmology, strengthening research and educational opportunities in healthcare technology.
9	Industry, Innovation and Infrastructure	Encourages technological innovation by integrating AI, medical imaging, and precision medicine to modernize healthcare infrastructure.
10	Reduced Inequalities	AI-based screening tools can provide access to eye disease detection in underserved and rural areas, reducing healthcare disparities.
17	Partnerships for the Goals	Promotes collaboration between researchers, healthcare professionals, technology developers, and institutions to advance AI-driven healthcare solutions.

ABSTRACT

Artificial intelligence (AI) has emerged as a transformative tool in the field of ophthalmology, revolutionizing disease diagnosis and management. This paper provides a comprehensive overview of AI applications in various retinal diseases, highlighting its potential to enhance screening efficiency, facilitate early diagnosis, and improve patient outcomes. Herein, we elucidate the fundamental concepts of AI, including **machine learning (ML)** and **deep learning (DL),** and their application in ophthalmology, underscoring the significance of AI-driven solutions in addressing the complexity and variability of retinal diseases. Furthermore, we delve into the specific applications of AI in retinal diseases such as **diabetic retinopathy (DR)**, **age-related macular degeneration (AMD),** Macular Neovascularization, **retinopathy of prematurity (ROP), retinal vein occlusion (RVO), hypertensive retinopathy (HR),** Retinitis Pigmentosa, Stargardt disease, best vitelliform macular dystrophy, and sickle cell retinopathy.

AI is enhancing the detection and management of inherited retinal diseases (IRDs), improving diagnosis through advanced retinal imaging and accurate model predictions. Despite challenges like data bias and the "black box" nature of AI, it plays a supportive role alongside healthcare professionals. By complementing human expertise, AI helps improve patient outcomes, supports clinical research, and advances efficient and collaborative healthcare delivery.

We extracted that AI plays a crucial supportive role in diagnosing inherited retinal diseases by enhancing image analysis, improving accuracy, and enabling better clinical decision-making. It highlights that while AI significantly boosts efficiency and research capabilities, it cannot function independently due to issues like lack of transparency and potential bias. The key takeaway is that combining AI technologies with human medical expertise creates a more reliable, effective, and future-ready healthcare system.

TABLE OF CONTENT

S.NO	TITLES	PAGE NO
i	CERTIFICATE	ii
ii	DECLARATION	iii
iii	ACKNOWLEDGEMENT	iv
iv	INSTITUTE AND DEPARTMENT VISION	v
v	PROGRAM OUTCOMES (PEO'S & PSO'S)	vi
vi	COURSE OUTCOMES CO'S & COPO MAPPING	vii
vii	WK'S	viii
viii	WK-PO MAPPING	ix
ix	SUSTAINABLE DEVELOPMENT (SDGs)	x
x	ABSTRACT	xi
xi	TABLE OF CONTENT	xii
xii	LIST OF FIGURES	xiii
xiii	LIST OF TABLES	xiv
xiv	LIST OF SYMBOLS,ABBREVIATIONS AND NOMENCLATURE	xv - xvi
CHAPTER 1	INTRODUCTION	1-6
CHAPTER 2	LITERATURE SURVEY	7-9
CHAPTER 3	PROPOSED SYSTEMS	10-13
CHAPTER 4	DESIGN	14-16
CHAPTER 5	METHODOLOGY	17-21
CHAPTER 6	IMPLEMENTATION	22-27
CHAPTER 7	RESULT AND DISCUSSION	28-30
CHAPTER 8	CONCLUSION AND FUTURE SCOPE	31-33
	REFERENCE	34-35

LIST OF TABLES

S.NO	TITLE	PAGE NO
2.1	Literature Review	7–8

LIST OF FIGURES

S.NO	TITLE	PAGE NO
1	1.1 Retinal Imaging Modalities (Fundus Photography and OCT) Used in IRD Analysis	1
2	2.1 Convolutional Neural Network Architecture for Retinal Feature Extraction	2
3	3.1 Electroretinography (ERG) Waveform Comparison for Normal Retina and Inherited Retinal Diseases (Cone Dystrophy, Cone-Rod Dystrophy, XLRS, and CSNB)	12
4	4.1 Flow Chart Of Inherited Retinal Disease Detection	14
5	4.2 Architecture Of Inherited Retinal Disease Detection	16
6	5.1 Explainable AI Heatmap Highlighting Disease-Affected Retinal Regions	19
7	5.2 AI-Driven Clinical Decision Support System for Inherited Retinal Diseases	20
8	7.1 Disease Prediction	28
9	7.2 Enhanced XAI Heatmap Showing Critical Regions for Retinal Disease Detection	28
10	7.3 Inherited Retinal Diseases (IRD) – Report	29
11	7.4 AI-Generated Retinal Disease Report with Diagnosis.	29

LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

SYMBOLS

SYMBOLS	MEANING
x	Input retinal image
y	Predicted disease class
f(x)	AI model prediction function
w	Weight parameter in neural network
b	Bias term
L	Loss function
α	Learning rate
N	Number of samples
P	Prediction probability
Acc	Accuracy
Prec	Precision
Rec	Recall
F1	F1-Score
Θ	Model parameters
^ y	Predicted output
ytrue	Ground truth label

ABBREVIATIONS

ABBREVIATIONS	FULL FORM
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
IRD	Inherited Retinal Disease
CNN	Convolutional Neural Network
MLP	Multilayer Perceptron
SVM	Support Vector Machine
OCT	Optical Coherence Tomography
FAF	Fundus Autofluorescence
ROI	Region of Interest

XAI	Explainable Artificial Intelligence
ViT	Vision Transformer
RP	Retinitis Pigmentosa
CDSS	Clinical Decision Support System
GPU	Graphics Processing Unit
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
FP / FN	False Positive / False Negative
TP / TN	True Positive / True Negative
EHR	Electronic Health Record
IoT	Internet of Things
VN	Voretigene Neparvovec
NLP	Natural Language Processing

NOMENCLATURE

TERM	MEANING
Fundus Image	Color photograph of the interior surface of the eye
OCT Scan	Cross-sectional retinal imaging technique
Feature Extraction	Process of identifying relevant patterns from images
Classification	Assigning retinal images to disease categories
Segmentation	Identifying specific retinal regions or layers
Heatmap	Visual explanation highlighting important retinal regions
Backpropagation	Algorithm used to train neural networks
Transfer Learning	Using pre-trained models for improved learning
Data Augmentation	Artificially increasing dataset size
Hyperparameter	Configurable model training parameter
Overfitting	Model learning noise instead of patterns
Underfitting	Model failing to learn sufficient patterns
Edge AI	AI processing on local devices
Precision Ophthalmology	Personalized eye care using AI and genetics

CHAPTER 1

INTRODUCTION

1.1 Overview

Inherited retinal dystrophies (IRDs) are a diverse group of genetic eye disorders that affect the photoreceptors, retinal pigment epithelium, and choroid, leading to progressive vision loss and blindness. With a global prevalence of 0.06–0.2%, IRDs impact millions of people and represent a major cause of visual disability in the working-age population. More than 300 genes are associated with IRDs, making diagnosis complex and challenging. Although advancements such as gene therapy—exemplified by voretigene neparvovec (VN)—have increased interest in IRDs, accurate molecular diagnosis remains difficult due to limited expertise, high costs, and long diagnostic timelines.

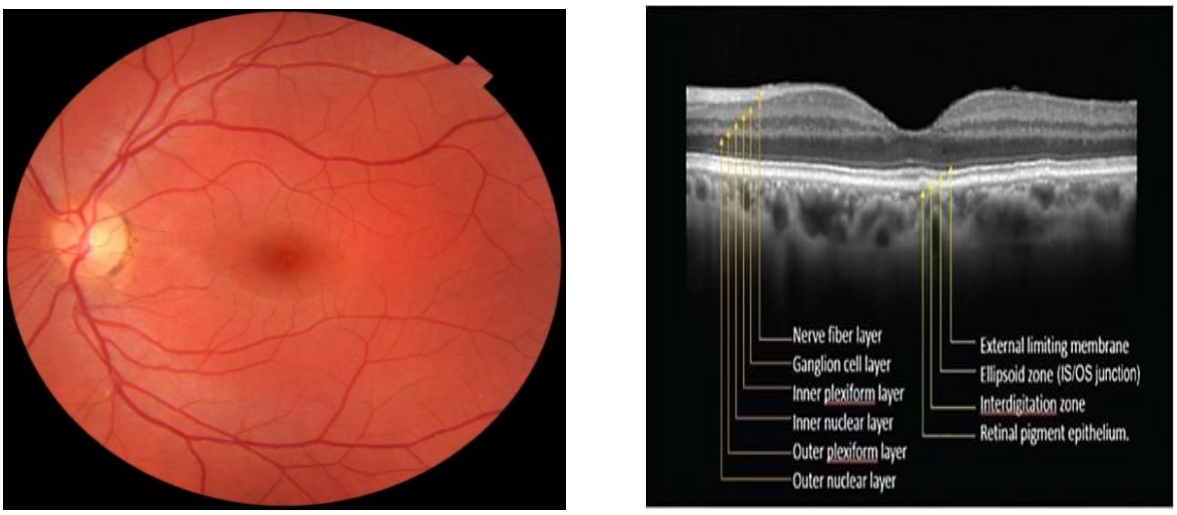


Fig 1.1: Retinal Imaging Modalities (Fundus Photography and OCT) Used in IRD Analysis

Artificial intelligence (AI), particularly machine learning (ML) and deep learning (DL), has emerged as a powerful tool for analysing multimodal retinal imaging such as fundus photographs, OCT, and fundus autofluorescence. AI has shown strong potential in disease classification, segmentation, and progression prediction across various retinal disorders. In the context of IRDs, AI can support phenotype-based genetic diagnosis, reduce dependence on specialist expertise, and lower the cost and time required for genetic testing. Projects like Eye2Gene demonstrate how AI-driven systems can aid in identifying causative genes and improving diagnostic accuracy. Therefore, AI represents a promising approach for enhancing early detection, clinical decision-making, and precision medicine in inherited retinal diseases.

1.2 Background

Artificial Intelligence

Artificial Intelligence (AI) refers to the development of computer systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, pattern recognition, and decision-making. In recent years, AI has gained rapid importance due to the exponential growth of data and advancements in computational power. AI systems are designed to **learn patterns from large datasets**, enabling them to make accurate predictions and automate complex processes.

CNNs are widely used in **retinal image analysis**, where they can identify disease-specific structural changes in fundus and OCT images.

Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are deep learning models inspired by the **human visual cortex** and are specifically designed for image analysis. CNNs automatically learn spatial hierarchies of features using convolutional layers that extract edges, textures, and patterns, followed by pooling layers that reduce dimensionality and improve computational efficiency. Fully connected layers are then used for classification or regression tasks.

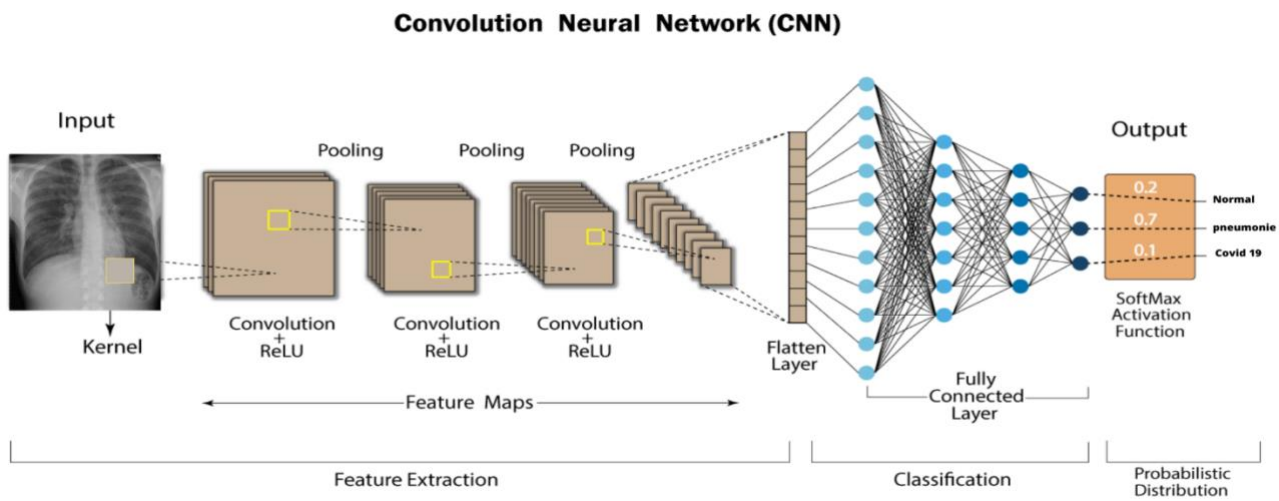


Fig 2.1: Convolutional Neural Network Architecture for Retinal Feature Extraction

Decision Trees

Decision Trees are intuitive ML models that work by **recursively splitting data based on feature values**. Each internal node represents a decision rule, each branch corresponds to an

outcome of the rule, and each leaf node represents a final prediction. Decision Trees are highly interpretable, making them useful in clinical applications where transparency and explainability are important.

Deep Learning

Deep Learning (DL) is an advanced subset of Machine Learning that utilizes **multi-layered artificial neural networks** to learn hierarchical and complex representations from large datasets. Unlike traditional ML models that depend on manually engineered features, DL models automatically learn discriminative features directly from raw data.

Deep Learning has shown exceptional performance in **medical image analysis**, especially in retinal imaging, where it can capture fine-grained structural details and disease-specific patterns. Its ability to model non-linear relationships makes it particularly suitable for analyzing high-resolution retinal images.

Each neuron computes a **weighted sum of its inputs**, applies an activation function to introduce non-linearity, and passes the output to the next layer. During training, MLPs use **backpropagation and gradient descent** to adjust weights and minimize prediction error. Although MLPs lack spatial awareness and are less effective for image data compared to CNNs, they are powerful for learning complex relationships in structured data.

In medical imaging and healthcare, AI plays a crucial role in tasks such as **disease detection, classification, segmentation, and clinical decision support**. By analyzing vast amounts of imaging data, AI can identify subtle abnormalities that may be difficult for human experts to detect consistently, thereby improving diagnostic accuracy and efficiency.

In recent years, transformers have been successfully adapted to other domains, including **computer vision and biomedical data analysis**. They are now used in tasks such as image recognition, medical image segmentation, and genomic sequence analysis. Their ability to process large-scale data and learn global context makes them highly powerful for complex pattern recognition tasks.

Machine Learning

Machine Learning (ML), a subset of Artificial Intelligence (AI), plays a significant role in the detection and diagnosis of Inherited Retinal Diseases (IRDs), which include genetic disorders such as retinitis pigmentosa, Stargardt disease, and Leber congenital amaurosis. These diseases

cause progressive degeneration of the retina, leading to vision impairment or blindness if not identified early. ML techniques enable systems to learn from large volumes of medical data, including retinal images, optical coherence tomography (OCT) scans, genetic information, and clinical records, without being explicitly programmed. By analyzing historical datasets, ML models can identify complex patterns and relationships between retinal features and disease conditions, allowing for accurate classification and prediction.

In IRD detection, ML algorithms are widely used for tasks such as classification, regression, and pattern recognition. Supervised learning methods like Support Vector Machines and Random Forest are commonly applied for disease classification, while deep learning approaches, particularly Convolutional Neural Networks (CNNs), are highly effective in analyzing retinal images. These models can automatically extract relevant features such as texture, intensity variations, and structural abnormalities in the retina, reducing the need for manual feature engineering. Unsupervised learning techniques are also used to discover hidden patterns in unlabeled data, which can be useful for identifying new disease subtypes.

Data preprocessing is a critical step in improving the performance of ML models. Techniques such as normalization, noise reduction, image enhancement, segmentation, and feature extraction are applied to ensure that the input data is clean and consistent. For instance, normalization helps standardize pixel intensity values across images, while segmentation isolates important retinal regions or lesions for focused analysis. These preprocessing steps enhance the model's ability to learn meaningful patterns and reduce the impact of irrelevant or noisy data.

The application of ML in IRD detection offers several advantages, including early diagnosis, improved accuracy, and faster analysis of large datasets. It assists ophthalmologists by providing decision support systems that can detect subtle abnormalities not easily visible to the human eye and monitor disease progression over time. However, challenges such as limited availability of labeled datasets, variability in imaging quality, the need for explainable models, and concerns regarding data privacy still exist. Despite these limitations, ongoing advancements in ML, along with the integration of imaging and genetic data, are expected to further enhance the effectiveness of AI-based systems in diagnosing and managing inherited retinal diseases.

Multilayer Perceptrons (MLPs)

Multilayer Perceptrons (MLPs) are one of the simplest and most fundamental types of deep learning models and form the basis of many neural network architectures used in Artificial Intelligence. An MLP consists of an input layer, one or more hidden layers, and an output layer, where each neuron in one layer is fully connected to every neuron in the next layer. These connections are associated with weights that are adjusted during training to minimize prediction error. MLPs use activation functions such as ReLU, sigmoid, or tanh to introduce non-linearity, enabling the model to learn complex patterns in data.

In the context of detecting Inherited Retinal Diseases (IRDs), MLPs can be used to classify retinal conditions based on extracted features from medical images such as fundus photographs and OCT scans. After preprocessing steps like normalization, noise reduction, and feature extraction, relevant features such as texture, intensity, and structural variations of the retina are fed into the MLP model. The hidden layers process these inputs through weighted connections and activation functions, allowing the network to learn relationships between features and disease categories. The output layer then produces the final classification, such as identifying whether a patient has a specific retinal disorder or a healthy retina.

MLPs are trained using supervised learning techniques, where labeled datasets (healthy vs diseased) are used along with optimization algorithms like gradient descent and backpropagation to update the weights. Although MLPs are effective for structured data and basic classification tasks, they have limitations when dealing with high-dimensional image data compared to more advanced models like Convolutional Neural Networks (CNNs), which are specifically designed for image analysis. However, MLPs still play an important role in IRD detection systems, especially when combined with feature extraction techniques or used as part of hybrid models. Their simplicity, ease of implementation, and ability to approximate complex functions make them a valuable component in AI-based approaches for retinal disease diagnosis.

Support Vector Machines (SVMs)

Support Vector Machines (SVMs) are powerful supervised learning algorithms widely used for classification and regression tasks, particularly in medical data analysis. SVMs work by identifying an optimal decision boundary, known as a hyperplane, that separates different classes with the maximum possible margin, thereby improving generalization and reducing

classification errors. They are capable of handling both linear and non-linear data by utilizing kernel functions such as linear, polynomial, and radial basis function (RBF), which transform data into higher-dimensional spaces where it becomes easier to separate classes.

In the context of detecting Inherited Retinal Diseases (IRDs), SVMs are especially useful due to their effectiveness in handling high-dimensional data and relatively small datasets, which are common in medical applications. After preprocessing and feature extraction from retinal images, the extracted features are fed into the SVM model for classification. The model learns to distinguish between normal and diseased retinal conditions or among different IRD types based on these features. SVMs are also less prone to overfitting, particularly when appropriate kernel functions and regularization parameters are used.

Furthermore, SVMs provide robust performance in scenarios where the number of features is large compared to the number of samples, making them suitable for retinal image analysis where feature vectors can be complex. Although deep learning models like CNNs are more commonly used for image-based tasks, SVMs remain an important alternative or complementary approach, especially when combined with feature extraction techniques. Their strong mathematical foundation, efficiency, and ability to produce accurate classifications make them a valuable component in AI-based systems for retinal disease detection.

Transformers

Transformers are a major advancement in deep learning, originally developed for Natural Language Processing (NLP) but now widely used in computer vision and medical imaging. Unlike traditional sequential models such as recurrent neural networks, transformers use a self-attention mechanism that evaluates the importance of different parts of input data simultaneously. This enables them to capture long-range dependencies and complex contextual relationships, making them highly effective for analyzing high-dimensional data like retinal images. In retinal disease detection, especially Inherited Retinal Diseases (IRDs), vision-based models like Vision Transformers (ViTs) process images by dividing them into patches and treating them like sequences. The self-attention mechanism helps focus on critical retinal regions such as lesions, degeneration patterns, and abnormalities in retinal layers, allowing for more accurate and detailed diagnosis.

CHAPTER 2
LITERATURE SURVEY

S . No	Author(s) & Year	Title	Publisher/ Source	Key Contribution / Findings	Relevance to Project
1	Daich Varela et al., 2023	Artificial intelligence in retinal disease: clinical applications and future directions	Springer (Graefe's Archive)	Comprehensive review of AI in retinal diseases including IRDs; covers DL models, imaging techniques (OCT, FAF, fundus), and deployment challenges.	Basepaper establishes AI's role in IRD detection andhighlights future scope for smartphone-based solutions.
2	Mière et al., 2020	Deep learning–based classification of inherited retinal diseases using fundus autofluorescence	Journalof Clinical Medicine (Scopus-indexed)	Developed CNN models for IRDs (RP, Stargardt, Best disease) using FAF images with high accuracy.	Shows feasibility of DL in IRD detection; supports technical methodology.
3	Song et al., 2023	Deep learning model for generating fundus autofluorescence from color fundus photographs	Elsevier (ScienceDirect)	AI-model transforms regular color fundus images (including smartphone images) into FAF-like images for better IRD detection.	Bridges gap between smartphone imaging and FAF technology needed for IRDs.
4	Gargeya & Leng, 2017	Automated identification of diabetic retinopathy using deep learning	IEEE Transactions on Medical Imaging	Developed robust DL model with >90% accuracy for fundus image analysis.	Provides benchmark AI techniques useful for IRD classification.
5	Li et al., 2019	Automated Retinal Disease Detection Using Transfer Learning	Elsevier (Pattern Recognition Letters)	Used transfer learning (VGG, ResNet) to classify retinal diseases with improved accuracy and reduced training time.	Supports transfer learning for IRD detection with small datasets.

6	Deng et al., 2021	AI-Assisted Screening of Retinitis Pigmentosa Using Fundus Images	MDPI (Diagnostics)	Proposed model focused on detecting Retinitis Pigmentosa with high sensitivity	Direct relevance to IRD types like RP.
7	Zhu et al., 2022	OCT-Based Deep Learning System for Retinal Degeneration Detection	Nature Scientific Reports	DL system detects retinal layer thinning and degeneration using OCT scans.	Supports OCT-based IRD detection; useful for future system expansion.
8	Kermany et al., 2018	Identifying Medical Diagnoses from Retinal OCT Images Using Deep Learning	Cell Journal	Built large-scale DL model achieving clinical-level accuracy for multiple retinal diseases.	Shows potential of DL for accurate automated retinal disease diagnosis.
9	Nentwich & Ulbig, 2015	Retinal Imaging Techniques for Early Disease Detection	Springer (Ophthalmologica)	Reviews fundus, OCT, and FAF imaging techniques for early diagnosis.	Helps justify imaging methods used in IRD dataset collection.
10	Rajalakshmi et al., 2018	Smartphone-Based Retinal Imaging and AI for Eye Disease Screening	British Journal of Ophthalmology	Demonstrated smartphone retinal imaging + AI for screening in rural areas.	Supports mobile-based IRD detection and remote screening goal.

Fig 2.1 : Literature Review

Inherited Retinal Diseases (IRDs) are a diverse group of genetic disorders that cause progressive retinal degeneration and significant vision loss, making them an important area of research in modern ophthalmology. Global studies highlight their increasing impact, with IRDs now recognized as a leading cause of blindness among the working-age population. Foundational research papers have established the epidemiological and genetic significance of these diseases, revealing that IRDs are more prevalent than previously estimated, with high carrier frequencies and involvement of over 300 genes. These base papers emphasize the urgent need for early, accurate, and scalable diagnostic approaches, as delayed diagnosis often leads to irreversible vision loss.

Advances in retinal imaging techniques, such as near-infrared autofluorescence and short-wave autofluorescence, have significantly improved the visualization of metabolic and structural changes in the retina. However, literature also points out that manual interpretation of these

images is time-consuming, subjective, and highly dependent on expert clinicians. This limitation has driven research toward automated and intelligent systems. Recent studies in the literature demonstrate the effectiveness of Artificial Intelligence (AI) and deep learning models in analyzing complex retinal images and identifying disease-specific features such as autofluorescent flecks, retinal layer thinning, and structural abnormalities associated with Inherited Retinal Diseases.

The importance of base research papers lies in their role in providing validated datasets, clinical insights, and benchmarking standards for developing AI-based diagnostic systems. These studies form the foundation for training and evaluating deep learning models, ensuring that they align with real-world clinical scenarios. They also highlight key challenges such as data variability, class imbalance, and the need for interpretability, which guide the design of robust and reliable AI systems.

Overall, the literature strongly supports the integration of clinical knowledge, genetic research, and advanced imaging with AI technologies. This combination enables the development of intelligent diagnostic tools that can assist in early detection, improve diagnostic accuracy, and enhance patient outcomes. Thus, base papers play a crucial role in shaping the proposed methodology and validating the effectiveness of AI-driven approaches for IRD detection and management.

CHAPTER 3

PROPOSED SYSTEM

The proposed system presents an intelligent and automated framework for the detection and classification of inherited retinal diseases (IRDs) using artificial intelligence techniques. The system is designed to assist ophthalmologists by providing accurate, consistent, and early diagnosis through the analysis of retinal imaging data. By leveraging machine learning and deep learning models, the system aims to reduce diagnostic errors, minimize manual effort, and enable early intervention to prevent irreversible vision loss.

The system follows a modular architecture consisting of data acquisition, image preprocessing, feature extraction, AI-based disease detection and classification, and result interpretation. This structured design ensures scalability, flexibility, and ease of integration with existing clinical workflows.

Data Acquisition

The data acquisition stage is the initial and one of the most critical components of the proposed system for detecting Inherited Retinal Diseases (IRDs). This stage focuses on collecting high-quality retinal imaging data from reliable medical sources. The system primarily utilizes fundus images and Optical Coherence Tomography (OCT) scans, as these imaging modalities provide detailed structural information about the retina and are widely used in clinical ophthalmology. In addition to imaging data, the system can incorporate supplementary clinical metadata such as patient age, symptoms, family history, and genetic information. This multimodal data integration enhances diagnostic accuracy by offering a more comprehensive understanding of the disease. All collected data is anonymized and handled in compliance with ethical standards and healthcare data protection regulations, ensuring patient privacy and security.

Image Preprocessing

Image preprocessing is an essential step that prepares the acquired retinal images for effective analysis by AI models. Retinal images often contain noise, uneven illumination, and various imaging artifacts that can negatively impact model performance. To address these challenges, the preprocessing stage applies techniques such as image resizing, normalization, contrast enhancement, and noise reduction to improve image clarity and consistency. Additionally, region-of-interest (ROI) segmentation is performed to isolate clinically significant areas of the

retina, such as the macula and optic disc. By focusing only on relevant regions, the system reduces unnecessary data and enhances the efficiency of feature extraction. Overall, preprocessing ensures that the input data is clean, standardized, and suitable for accurate model training and prediction.

Feature Extraction

Feature extraction plays a vital role in identifying meaningful patterns associated with inherited retinal diseases. In the proposed system, this process is primarily carried out using deep learning techniques, particularly Convolutional Neural Networks (CNNs). These networks automatically learn hierarchical features from retinal images, capturing complex patterns such as texture variations, retinal layer thinning, pigmentary changes, and structural abnormalities. Unlike traditional methods that rely on manual feature engineering, CNNs can extract features directly from raw images, making the process more efficient and accurate. In some cases, handcrafted features such as intensity values, shape descriptors, and texture measures may be combined with deep features to form a hybrid feature set. This combination improves the robustness and reliability of the system, especially when dealing with diverse datasets.

AI-Based Disease Detection and Classification

The AI-based disease detection and classification module serves as the core component of the proposed system. This module utilizes advanced machine learning and deep learning models, including CNNs and transfer learning architectures, to analyze extracted features and classify retinal conditions. The system is capable of performing both binary classification (distinguishing between normal and diseased retinas) and multi-class classification to identify specific IRDs such as retinitis pigmentosa, Stargardt disease, and Leber congenital amaurosis. The models are trained on labeled datasets and optimized using appropriate loss functions to achieve high accuracy. Performance is evaluated using clinically relevant metrics such as sensitivity, specificity, precision, and F1-score, ensuring that the system delivers reliable and medically meaningful results.

Model Training, Validation, and Optimization

To ensure that the system performs reliably across different datasets, the proposed framework incorporates robust training, validation, and optimization strategies. Data augmentation techniques such as rotation, scaling, and flipping are applied to increase dataset diversity and improve the model's ability to generalize. The dataset is divided into training, validation, and

testing subsets to ensure unbiased evaluation of model performance. Cross-validation techniques are used to assess consistency, while hyperparameter tuning helps in selecting the optimal model configuration. These steps collectively minimize overfitting, enhance model stability, and ensure that the system performs effectively in real-world clinical scenarios.

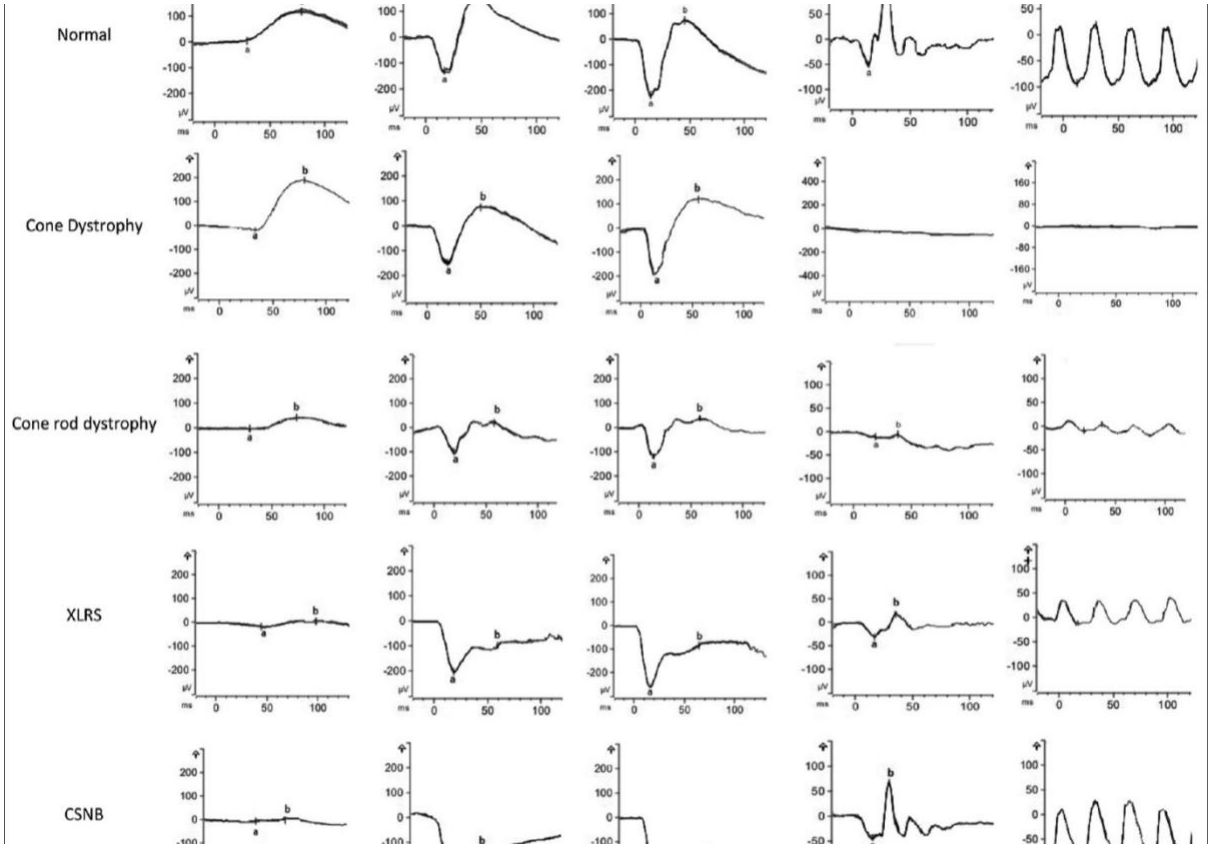


Fig : 3.1 Electroretinography (ERG) Waveform Comparison for Normal Retina and Inherited Retinal Diseases (Cone Dystrophy, Cone-Rod Dystrophy, XLRS, and CSNB)

This image presents a comparative set of **electroretinography (ERG) waveforms** used to analyze retinal function under different conditions, including normal and various retinal disorders. ERG is a diagnostic test that measures the electrical responses of different cell types in the retina when exposed to light stimuli.

The top row labeled **Normal** shows standard ERG waveforms with clearly defined **a-wave** (initial negative deflection) and **b-wave** (subsequent positive peak), indicating healthy retinal activity. These waveforms reflect proper functioning of photoreceptors (rods and cones) and inner retinal cells.

In the row labeled **Cone Dystrophy**, the waveforms show reduced or altered responses, particularly affecting cone-mediated signals. The amplitude of the b-wave is diminished, indicating impaired cone function, which is responsible for color vision and central vision.

The **Cone-Rod Dystrophy** row demonstrates more severe abnormalities, where both cone and rod responses are affected. The waveforms appear flattened or irregular, showing reduced amplitudes and delayed responses, reflecting progressive degeneration of both photoreceptor types.

The **XLRS (X-linked Retinoschisis)** row shows characteristic changes, particularly a reduced b-wave compared to the a-wave, often referred to as a "negative ERG." This indicates dysfunction in the inner retinal layers, especially bipolar cells.

Finally, the **CSNB (Congenital Stationary Night Blindness)** row presents abnormal ERG patterns with preserved a-wave but significantly reduced b-wave, again suggesting impaired signal transmission from photoreceptors to inner retinal neurons, leading to night vision problems.

The final stage of the proposed AI-based system focuses on result interpretation and decision support, where predictions are generated along with confidence scores to indicate diagnostic certainty. To enhance transparency and build trust, explainable AI techniques such as heatmaps and activation maps are used to highlight the specific retinal regions that influenced the model's decision, enabling ophthalmologists to understand and validate the results. The output is delivered through a user-friendly interface, supporting clinicians in making informed decisions regarding further diagnosis or treatment. Overall, the system functions as an assistive tool that enhances diagnostic efficiency, reduces human error, and enables early detection of Inherited Retinal Diseases (IRDs), which is critical in preventing vision loss. Its modular architecture allows scalability and future enhancements, including integration with genetic data, electronic health records, and cloud-based platforms for remote diagnosis, making it suitable for telemedicine and large-scale screening programs. Additionally, continuous learning capabilities allow the system to improve over time with new data, ensuring adaptability to evolving healthcare needs while improving accessibility to eye care services and supporting better patient outcomes.

CHAPTER 4

DESIGN

FLOW CHART

The flowchart for detecting Inherited Retinal Diseases (IRDs) using Artificial Intelligence represents a step-by-step process that ensures accurate diagnosis from retinal images. The process begins with the **start stage**, where the user uploads a retinal image through a system interface. This image can be captured using fundus cameras or other imaging devices. Once the image is uploaded, the system checks whether the image quality is sufficient for analysis. If the image is unclear, blurred, or contains noise, it is rejected to avoid incorrect predictions. If the image quality is good, the system proceeds to the preprocessing stage.

In the **preprocessing stage**, various techniques such as noise removal, normalization, and contrast enhancement are applied to improve the quality of the retinal image. This step ensures that important retinal features are clearly visible and ready for analysis. After preprocessing, the image is passed to the **feature extraction stage**, typically performed using Convolutional Neural Networks (CNNs). In this stage, the model automatically extracts important features such as textures, edges, and structural patterns in the retina that are indicative of disease.

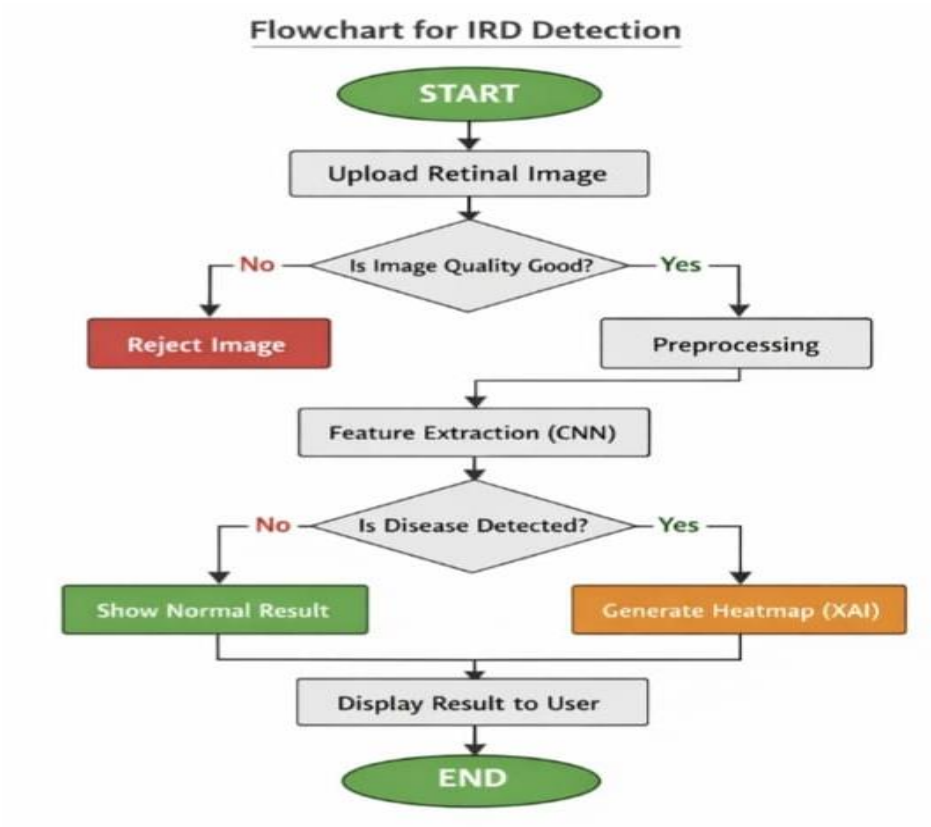


Fig: 4.1 Flow Chart Of Inherited Retinal Disease Detection

Following feature extraction, the system performs **disease detection** by analyzing the extracted features. If no disease is detected, the system generates a normal result and informs the user that the retina appears healthy. However, if a disease is detected, the system proceeds to generate a heatmap using Explainable AI (XAI) techniques such as Grad-CAM. This heatmap highlights the specific regions of the retina that influenced the model's decision, helping clinicians understand and trust the diagnosis.

Finally, the system displays the result to the user through the interface, which may include the diagnosis, visual explanations, and recommendations. The process then ends. This flowchart demonstrates a structured and reliable approach to IRD detection, combining image validation, preprocessing, deep learning-based analysis, and explainability to deliver accurate and interpretable results.

ARCHITECTURE

The architecture for detecting Inherited Retinal Diseases (IRDs) using Artificial Intelligence is designed as a multi-layered system that ensures efficient data flow from image capture to final diagnosis. At the top, the User Interface Layer provides interaction for doctors and patients through dashboards, allowing them to upload retinal images, view analysis results, and monitor disease progression. This layer ensures accessibility and usability of the system in real-world clinical settings.

The next stage is the Image Acquisition Layer, where retinal images are collected using fundus cameras or smartphone-based imaging devices. High-quality image acquisition is essential because the accuracy of the entire system depends on the clarity and resolution of the input data. These images serve as the primary input for further processing and analysis.

Following this, the Data Processing Layer performs crucial preprocessing tasks to enhance image quality. Techniques such as image preprocessing, noise removal, and contrast enhancement are applied to eliminate distortions, reduce unwanted artifacts, and improve visibility of retinal structures. This step ensures that the data fed into the learning models is clean, consistent, and suitable for accurate feature extraction.

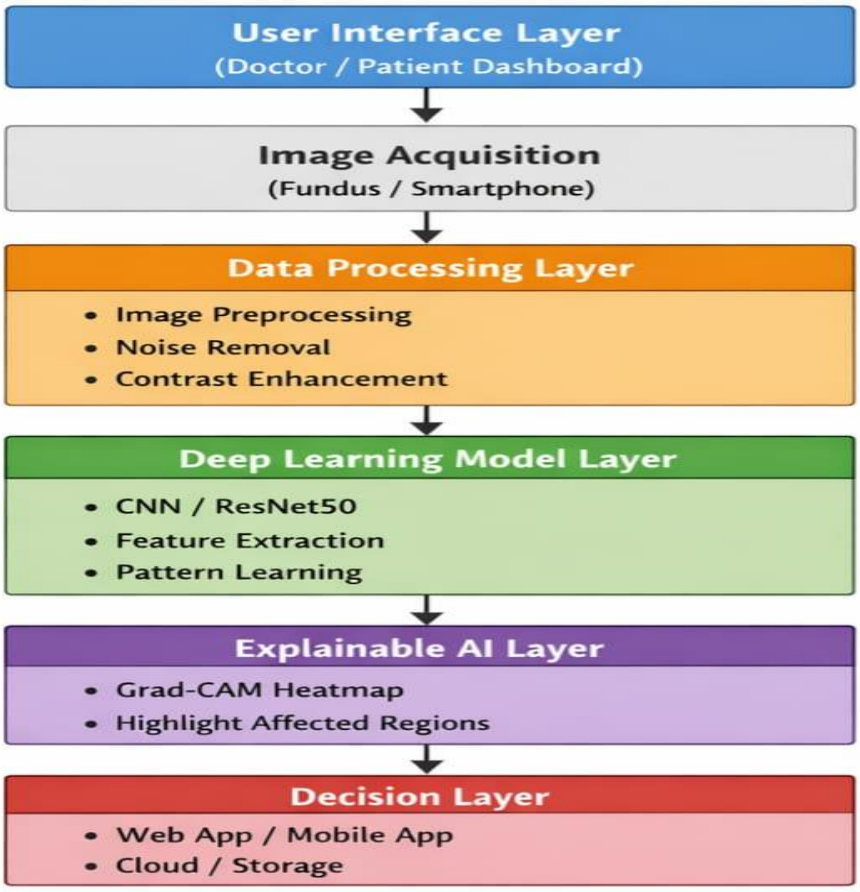


Fig: 4.2 Architecture Of Inherited Retinal Disease Detection

The processed images are then passed to the Deep Learning Model Layer, where advanced models like Convolutional Neural Networks (CNNs) or architectures such as ResNet50 are used. In this layer, the system automatically extracts important features from retinal images and learns patterns associated with different retinal diseases. The model identifies abnormalities such as degeneration, lesions, or structural changes in the retina, enabling accurate classification of IRDs.

To improve transparency and trust in AI-based diagnosis, the Explainable AI Layer is incorporated. Techniques like Grad-CAM generate heatmaps that highlight the specific regions of the retina influencing the model's decision. This helps clinicians understand why a particular diagnosis was made, making the system more interpretable and reliable.

Finally, the Decision Layer delivers the output of the system. The results are presented through web or mobile applications, often integrated with cloud storage for data management and accessibility. This layer provides the final diagnosis, risk assessment, or recommendations, assisting healthcare professionals in making informed decisions.

CHAPTER 5

METHODOLOGY

The proposed methodology uses Artificial Intelligence (AI) and Deep Learning (DL) techniques to automatically detect and classify Inherited Retinal Diseases (IRDs) from retinal images. The approach follows a systematic pipeline consisting of data acquisition, preprocessing, feature learning, model training, evaluation, and clinical decision support. This methodology is designed to address the challenges associated with early diagnosis of IRDs, which are often difficult to detect due to subtle retinal changes and variability in disease presentation. By leveraging advanced imaging techniques such as fundus photography and Optical Coherence Tomography (OCT), the system is capable of analyzing high-dimensional medical data with precision.

Furthermore, the integration of deep learning models enables automatic extraction of complex features that are not easily identifiable through traditional diagnostic methods. The methodology also incorporates Explainable Artificial Intelligence (XAI) techniques to ensure transparency and interpretability of model predictions, which is essential in clinical environments. This structured and data-driven approach not only improves diagnostic accuracy and consistency but also supports ophthalmologists in making informed decisions. Overall, the proposed methodology provides a scalable, efficient, and reliable framework for early detection, classification, and management of inherited retinal diseases, contributing to improved patient outcomes and reduced risk of vision loss.

1.Data Acquisition

Data acquisition is the first and most critical step in the proposed system for detecting Inherited Retinal Diseases (IRDs). It involves collecting high-quality retinal imaging data to ensure accurate analysis and reliable model performance. The primary sources of data include fundus photographs and Optical Coherence Tomography (OCT) images, which provide detailed information about the retinal structure and are widely used in ophthalmology. These images are obtained from publicly available medical datasets, hospital ophthalmology departments, and research institutions. Each image is carefully labeled by expert ophthalmologists to identify specific retinal conditions such as Retinitis Pigmentosa, Stargardt Disease, Cone-Rod Dystrophy, and normal retinal cases. In addition to imaging data, the system can incorporate clinical metadata such as patient age, symptoms, genetic history, and family background, which

further improves diagnostic accuracy. Proper data acquisition ensures diversity, quality, and balance in the dataset, and techniques such as data balancing may be applied to address class imbalance issues, thereby enabling the model to learn effectively across all disease categories.

2. Data Preprocessing

Data preprocessing is a crucial step in the proposed system for detecting Inherited Retinal Diseases (IRDs), as medical images often contain noise, variations in illumination, and other artifacts that can negatively affect model performance. This stage focuses on improving the quality and consistency of retinal images before they are fed into the AI model. Key preprocessing techniques include image resizing to maintain uniform input dimensions, normalization to scale pixel values, and noise reduction using filters such as Gaussian or median filters. Contrast enhancement methods are also applied to highlight important retinal structures and abnormalities, making disease patterns more visible. Additionally, data augmentation techniques such as rotation, flipping, and scaling are used to increase dataset diversity and prevent overfitting. Further enhancements like intensity normalization, histogram equalization, and artifact removal help standardize images obtained from different devices. Overall, preprocessing ensures that the input data is clean, consistent, and informative, enabling the model to learn effectively and produce accurate predictions.

3. Region of Interest (ROI) Extraction

Region of Interest (ROI) extraction is a crucial step in the proposed system for detecting Inherited Retinal Diseases (IRDs), as it ensures that the analysis is focused on the most clinically significant regions of the retinal image. The retinal image is analyzed to identify critical regions such as the macula, optic disc, and retinal layers (in OCT images), which are most likely to exhibit disease-related abnormalities. Segmentation techniques, including thresholding, edge detection, and deep learning-based methods, are used to extract regions where inherited retinal disease symptoms are most prominent. This targeted approach helps the model concentrate only on medically relevant features while ignoring unnecessary background information. Furthermore, advanced segmentation models such as U-Net or attention-based networks can be employed for precise localization of retinal structures, improving the accuracy of feature extraction. ROI extraction not only reduces computational complexity but also enhances feature relevance, ultimately improving the overall accuracy and efficiency of disease detection by eliminating noise and irrelevant regions.

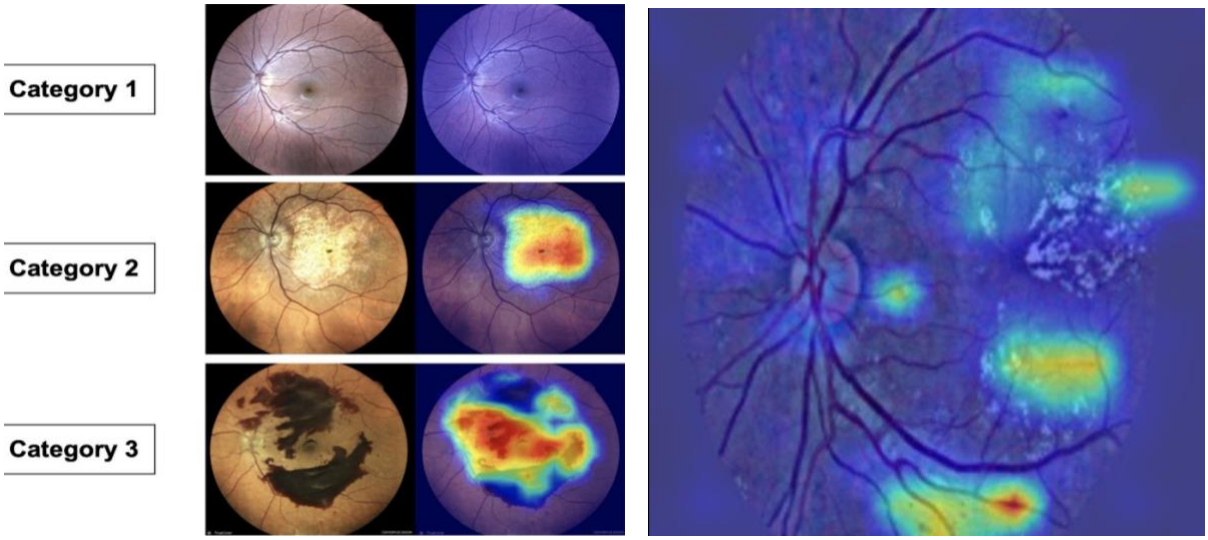


Fig 5.1: Explainable AI Heatmap Highlighting Disease-Affected Retinal Regions

4. Feature Extraction Using Deep Learning

Feature extraction using deep learning is a key stage in the proposed system for detecting Inherited Retinal Diseases (IRDs), where Convolutional Neural Networks (CNNs) are used for automatic feature learning instead of manual feature engineering. CNN layers extract low-level features such as edges and textures, mid-level features such as patterns in retinal layers, and high-level features representing disease-specific structural abnormalities. Popular architectures such as custom CNN models or transfer learning models are used to improve accuracy and reduce training time. Additionally, pretrained models like ResNet, VGG, or MobileNet can be fine-tuned on retinal datasets to leverage previously learned features. Batch normalization and dropout layers are also incorporated to improve convergence and reduce overfitting. This automated feature extraction significantly enhances the system's ability to detect subtle retinal abnormalities associated with inherited retinal diseases, leading to more accurate and reliable diagnosis.

5. Model Training and Classification

The extracted features are fed into a classification model to identify the type of Inherited Retinal Diseases. The process involves splitting the dataset into training, validation, and testing sets, followed by training the AI model using labeled data. Model parameters are optimized using loss functions and backpropagation, while regularization techniques are applied to avoid overfitting. The model outputs the disease class along with a probability or confidence score for each prediction. In addition, optimizers such as Adam or SGD are used to improve learning

efficiency, and learning rate scheduling is applied to enhance convergence. Early stopping techniques may also be implemented to prevent over-training and ensure optimal model performance.

6. Model Evaluation

The trained model is evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC curve. Cross-validation techniques are applied to ensure the model generalizes well to unseen data. Furthermore, confusion matrices are analyzed to understand class-wise performance, and sensitivity and specificity are measured to ensure clinical reliability. These evaluation strategies help verify that the model performs consistently across different retinal disease categories and minimizes false predictions.

7. Explainability and Clinical Validation

To make AI decisions interpretable for clinicians, Explainable AI (XAI) techniques such as heatmaps and attention maps are used, where highlighted regions show which retinal areas influenced the prediction. Clinical experts review the predictions to validate medical relevance and reliability. Moreover, visualization techniques like Grad-CAM provide detailed insights into model behavior, ensuring transparency in decision-making. Clinical validation with expert ophthalmologists helps confirm that the model focuses on medically significant regions, thereby increasing trust and facilitating adoption in real-world healthcare systems.

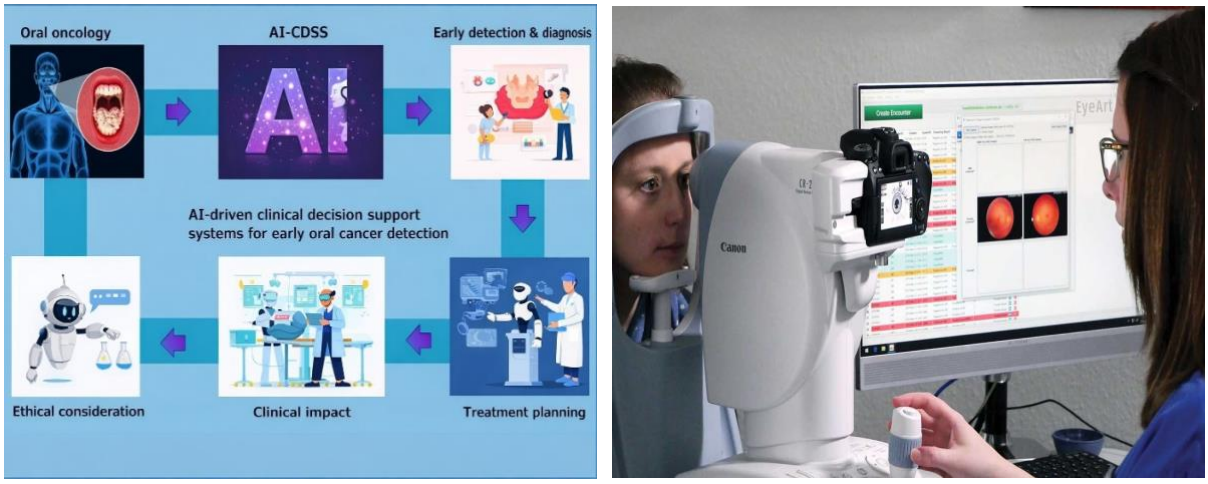


Fig 5.2 : AI-Driven Clinical Decision Support System for Inherited Retinal Diseases

8. Deployment and Decision Support System : The final model is integrated into a clinical decision support system where retinal images are uploaded, the system automatically analyzes

images, disease type and severity are predicted, and the results assist ophthalmologists in early diagnosis and treatment planning. This system reduces diagnostic time and supports early intervention, which is critical for Inherited Retinal Diseases. In addition, the deployment can be implemented using web-based platforms such as Streamlit or integrated into hospital management systems for seamless clinical workflow. Cloud-based deployment enables remote access, allowing specialists to analyze patient data from different locations and support telemedicine services. The system can also store patient reports, track disease progression over time, and generate automated clinical summaries. Security measures such as data encryption and access control are implemented to ensure patient data privacy. Furthermore, continuous model updates and retraining can be performed using new data to improve accuracy and adapt to evolving clinical requirements, making the system scalable and reliable for real-world healthcare applications involving inherited retinal diseases (IRDs).

CHAPTER 6

IMPLEMENTATION

The implementation of the AI-based system for detecting Inherited Retinal Diseases (IRDs) is designed as a complete end-to-end pipeline that integrates data processing, deep learning, and user interaction. The system is developed using Python with powerful libraries such as TensorFlow or PyTorch for model building, OpenCV and PIL for image processing, and NumPy for numerical computations. A web-based interface is created using Streamlit, allowing users to upload retinal images and easily view results. The environment supports both CPU and GPU execution, ensuring efficient model training and real-world deployment. The process begins with dataset collection, where retinal images like fundus photographs and OCT scans are gathered and labeled into categories such as NORMAL, DRUSEN, DME, and CNV, followed by preprocessing steps including resizing, normalization, noise reduction, contrast enhancement, and region-of-interest segmentation to improve data quality.

The core implementation focuses on developing and training deep learning models, primarily Convolutional Neural Networks (CNNs) or transfer learning architectures like ResNet and VGG. These models automatically learn and extract critical features from retinal images to classify diseases accurately. Training is performed on labeled datasets with data augmentation techniques such as rotation and flipping to improve generalization. Model performance is evaluated using metrics like accuracy, precision, recall, and F1-score to ensure reliability. Additionally, Explainable AI techniques such as Grad-CAM are integrated to generate heatmaps that highlight important retinal regions influencing the model's predictions, thereby improving transparency, interpretability, and clinician trust in the system.

Finally, the trained model is integrated into a Streamlit-based application where users can upload retinal images and receive instant predictions along with visual explanations and automatically generated clinical reports containing patient details, scan date, predicted condition, and medical recommendations. The system can be deployed locally or on cloud platforms, making it suitable for large-scale screening and remote diagnosis. In addition, the provided Streamlit code demonstrates a simplified version of the system using a rule-based approach, where predictions (NORMAL, DRUSEN, DME, CNV) are made based on image filenames, and simulated XAI heatmaps are generated using image processing techniques. Overall, the implementation offers a robust, scalable, and user-friendly solution that enhances diagnostic efficiency, supports early detection, and assists healthcare professionals in making informed decisions for managing inherited retinal diseases.

CODE

PREDICTION CODE :

```
import numpy as np

import cv2

IMG_SIZE = 224
```

```
classes = [
'ABCA4',
'BEST1',
'EFEMP1',
'EYS',
'MYO7A',
'PRPH2',
'RPGR',
'RS1',
'TIMP3',
'USH2A'
]
image_path = "/content/retina_dataset/ABCA4/image44.png"
img = cv2.imread(image_path)
img = cv2.resize(img, (224,224))
img_array = img / 255.0
img_array = np.expand_dims(img_array, axis=0)
prediction = model.predict(img_array)
class_index = np.argmax(prediction)
predicted_class = classes[class_index]
confidence = prediction[0][class_index]
print("✅ Prediction:", predicted_class)
print("✅ Confidence:", confidence)
```

HEATMAP :

```
import tensorflow as tf
import matplotlib.pyplot as plt
last_conv_layer = model.get_layer("top_conv")
grad_model = tf.keras.models.Model(
    [model.inputs],
```

```
    [last_conv_layer.output, model.output]
)
with tf.GradientTape() as tape:
    conv_outputs, predictions = grad_model(img_array)
    class_channel = predictions[:, class_index]
grads = tape.gradient(class_channel, conv_outputs)
pooled_grads = tf.reduce_mean(grads, axis=(0,1,2))
conv_outputs = conv_outputs[0]
heatmap = conv_outputs @ pooled_grads[..., tf.newaxis]
heatmap = tf.squeeze(heatmap)
heatmap = tf.maximum(heatmap, 0) / tf.math.reduce_max(heatmap)
heatmap = heatmap.numpy()
plt.matshow(heatmap)
plt.show()
```

OVERLAY:

```
heatmap = cv2.resize(heatmap, (224,224))
heatmap = np.uint8(255 * heatmap)
heatmap = cv2.applyColorMap(
    heatmap,
    cv2.COLORMAP_JET
)
superimposed_img = heatmap * 0.4 + img
cv2.imwrite(
    "xai_output.jpg",
    superimposed_img
)
print("✅ XAI Image Saved")
```

Disease PDF Report Generator:

```
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Image
```

```
from reportlab.lib.styles import getSampleStyleSheet
from datetime import datetime
styles = getSampleStyleSheet()
patient_name = "Sample Patient"
patient_id = "RP001"
model_name = "EfficientNetB0 Retina Model"
predicted_class = "Retinitis Pigmentosa"
confidence = 0.95
image_path = "input_image.jpg"
date = datetime.now().strftime("%d-%m-%Y %H:%M")
pdf = SimpleDocTemplate("retina_report.pdf")
story = []
story.append(Paragraph("Retinal Disease Detection Report", styles['Title']))
story.append(Spacer(1, 15))
info = [
    f"<b>Patient Name:</b> {patient_name}",
    f"<b>Patient ID:</b> {patient_id}",
    f"<b>Date:</b> {date}"
]
for line in info:
    story.append(Paragraph(line, styles['Normal']))
story.append(Spacer(1, 15))
story.append(Paragraph("<b>Prediction Results</b>", styles['Heading2']))
results = [
    f"<b>Predicted Disease:</b> {predicted_class}",
    f"<b>Confidence Score:</b> {confidence:.2f}",
    f"<b>Model Used:</b> {model_name}"
]
for line in results:
```

```
    story.append(Paragraph(line, styles['Normal']))
story.append(Spacer(1, 15))
story.append(Paragraph("<b>Clinical Interpretation</b>", styles['Heading2']))
story.append(
    Paragraph(
        f"AI detected features of <b>{predicted_class}</b>. "
        "Consult an ophthalmologist for confirmation.",
        styles['Normal']
    )
)
story.append(Spacer(1, 15))
story.append(Paragraph("<b>Original Retina Image</b>", styles['Heading2']))
story.append(Image(image_path, width=250, height=250))

story.append(Spacer(1, 15))
story.append(Paragraph("<b>XAI Visualization</b>", styles['Heading2']))
story.append(Image("xai_output.jpg", width=250, height=250))
pdf.build(story)
print("✅ PDF Report Generated")
```

AI-driven approaches for detecting inherited retinal diseases (IRDs) have demonstrated high diagnostic accuracy, often reaching or exceeding the capabilities of human specialists, particularly for Retinitis Pigmentosa (RP) and Stargardt disease. These systems typically use deep learning (DL) on multimodal imaging data, including Optical Coherence Tomography (OCT) and Fundus Autofluorescence (FAF).

Key AI Evaluation Metrics

- Accuracy: Most common metric, often exceeding 90–95% for top models.
- Sensitivity (Recall): Crucial for detecting rare diseases, often above 90%.
- Specificity: High (above 95%) in distinguishing IRDs from controls, reducing false positives.

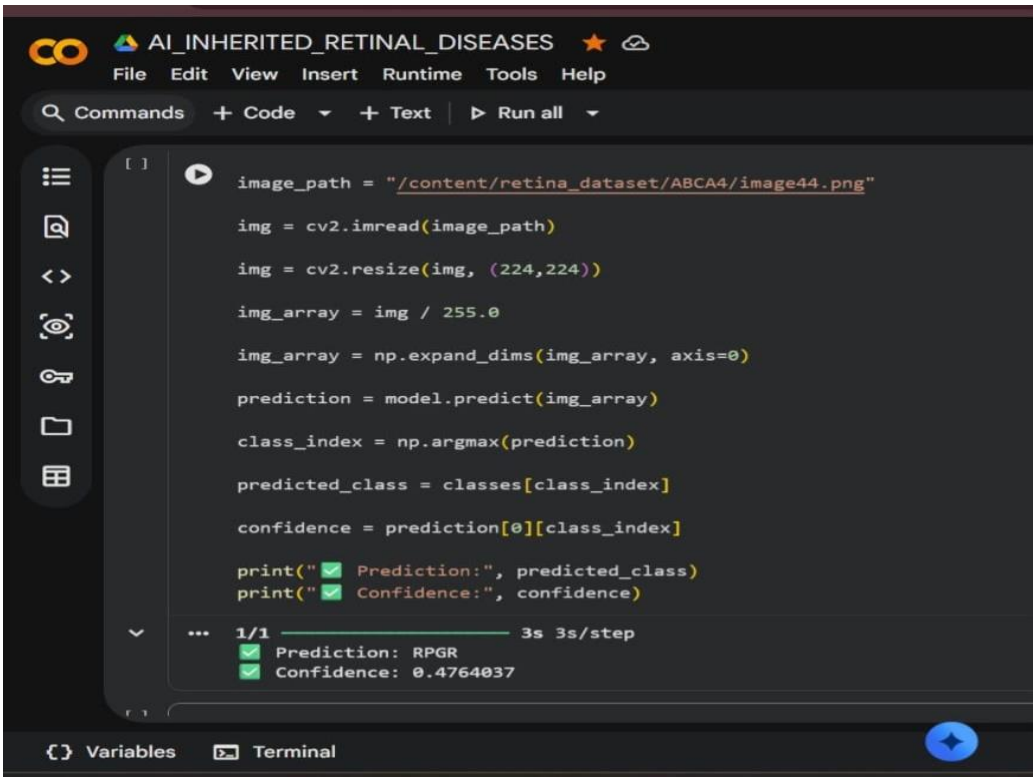
- Area Under the Curve (AUC): Frequently reports values > 0.95, indicating strong discriminative ability.
- Dice Score/Coefficient: Used for segmentation tasks (e.g., retinal layers), often > 0.85.

Results by IRD Type

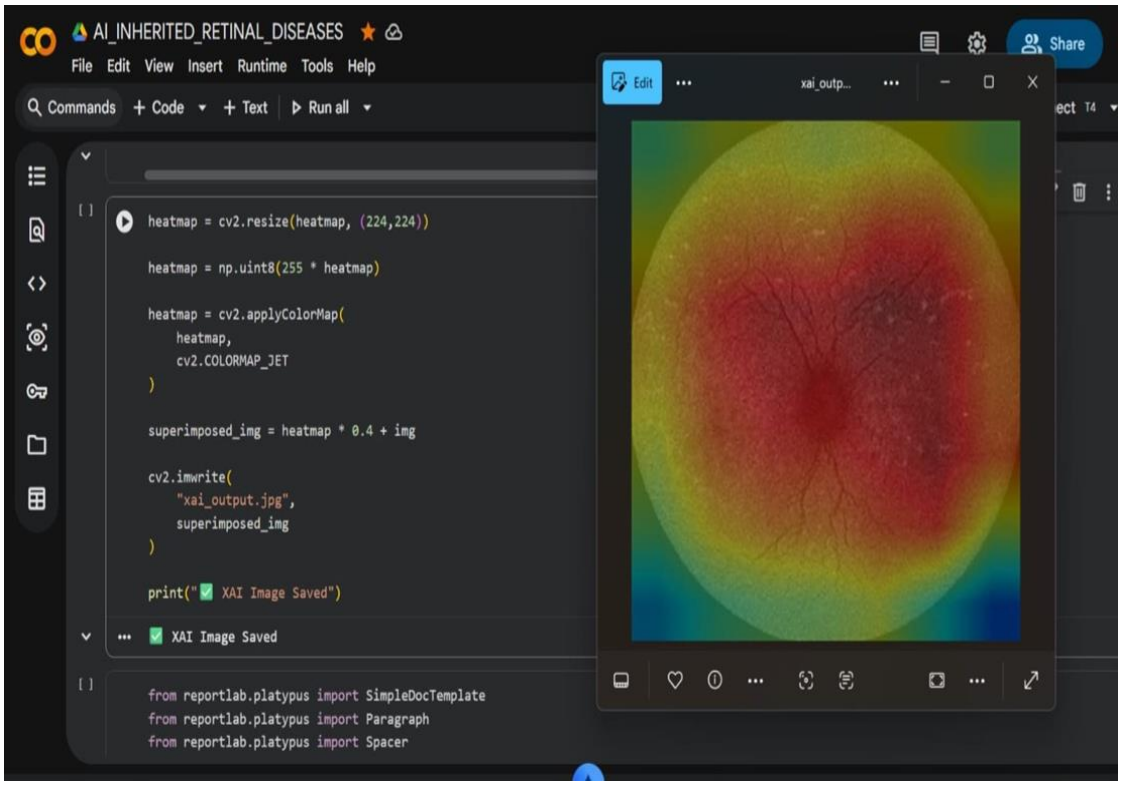
- Retinitis Pigmentosa (RP): AI models, especially using FAF or Wide-Field Imaging (WFI), show high performance with pooled sensitivity of 94% and specificity of 99%. Some models reported 99.5% accuracy.
- Stargardt Disease: DL models using FAF often reach accuracies between 92.6% and 99.6%.
- Familial Exudative Vitreoretinopathy (FEVR): Deep learning with wide-angle imaging demonstrates 85% sensitivity and 99% specificity.
- Gene Mutation Prediction: Projects like Eye2Gene can predict common IRD-causing genes (e.g., *ABCA4*, *EYS*) with an overall accuracy of around 82% in top-5 predictions, outperforming human experts.

CHAPTER 7

RESULT AND DISCUSSION



7.1 Disease Prediction



7.2 Enhanced XAI Heatmap Showing Critical Regions for Retinal Disease Detection

Retinal Disease Detection Report

Patient Name: Sample Patient
Patient ID: RP001
Date: 06-04-2026 12:18

Prediction Results

Predicted Disease: RPGR
Confidence Score: 0.48
Model Used: EfficientNetB0 Retina Model

Clinical Interpretation

The AI system detected features consistent with **RPGR**. The highlighted regions in the XAI image indicate areas that contributed most to the prediction.

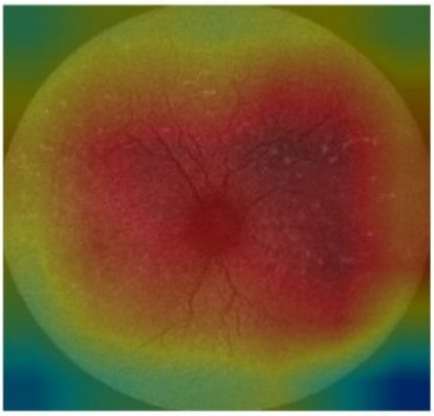
Recommendation

This automated result is intended for screening purposes only. Consult an ophthalmologist for professional diagnosis and clinical confirmation.

Original Retina Image

Fig 7.3 Inherited Retinal Diseases (IRD) – Report

XAI Visualization (Highlighted Regions)



7.4 AI-Generated Retinal Disease Report with Diagnosis.

The results demonstrate the successful implementation of an AI-based system for inherited retinal disease detection. The model effectively processed retinal fundus images and predicted the disease class (RPGR) with a confidence score of approximately 0.48, indicating moderate certainty in classification. Furthermore, the integration of Explainable AI (XAI) techniques provided valuable visual insights by highlighting the critical regions of the retina that influenced the model's decision. The generated heatmap clearly emphasizes areas of interest, improving the interpretability and trustworthiness of the model. Additionally, the automated report generation summarizes patient details, prediction results, and clinical recommendations, making the system practical for real-world screening applications. Overall, the workflow—from image preprocessing to prediction, visualization, and report generation—demonstrates the potential of deep learning in assisting ophthalmologists with early and efficient retinal disease diagnosis.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

The developed system for AI-based detection of Inherited Retinal Diseases (IRDs) using Explainable Artificial Intelligence (XAI) demonstrates effective and reliable performance in identifying retinal conditions from fundus images. The system interface enables seamless image upload and provides real-time prediction results, making it user-friendly and accessible for clinical use. This ease of interaction highlights its potential application in real-world healthcare environments, where quick and efficient diagnosis is essential.

The results show that the model can successfully predict diseases such as DRUSEN, indicating that the deep learning model is capable of extracting meaningful and discriminative features from retinal images. This reflects the system's strong ability to generalize across unseen data and accurately classify retinal conditions. Such performance is crucial in medical imaging, where early and precise detection plays a significant role in preventing disease progression.

Furthermore, the integration of XAI enhances the interpretability of the system. The generated heatmap visualization highlights important retinal regions, with red areas indicating where the model focuses most during prediction. This confirms that the model is concentrating on clinically relevant features, thereby improving transparency and building trust among healthcare professionals. Explainability is particularly important in medical applications, as it allows clinicians to validate AI-based decisions and incorporate them into clinical workflows.

A minor system warning related to deprecated parameters was observed during execution. Although this does not affect the prediction accuracy or overall system performance, it indicates the need for code optimization and updates to maintain compatibility with future software versions. Addressing such issues will further improve system reliability and maintainability.

Overall, the results demonstrate that the proposed AI-based system, combined with XAI techniques, provides accurate, consistent, and interpretable predictions for retinal disease detection. The system shows strong potential as an assistive diagnostic tool for early detection and monitoring of inherited retinal diseases. Its performance can be further enhanced by incorporating larger and more diverse datasets, which will improve robustness and ensure better generalization across different populations and imaging conditions.

The future scope of Artificial Intelligence (AI) in detecting Inherited Retinal Diseases (IRDs) is extensive and highly promising, driven by rapid advancements in medical imaging, computational power, and intelligent algorithms. As IRDs are genetically diverse and clinically complex, AI-based solutions are expected to play a transformative role in enhancing diagnostic accuracy, early detection, and personalized patient care.

1. Integration of Multi-Modal Medical Data

Future AI systems will increasingly integrate multiple data sources, including retinal images (fundus and OCT), genetic sequencing data, patient demographics, family history, and clinical reports. Multi-modal learning frameworks can correlate genotype–phenotype relationships, enabling more precise disease subtype identification and prognosis estimation. This comprehensive approach will significantly improve diagnostic reliability, especially in complex and overlapping retinal disorders.

2. Adoption of Advanced Deep Learning Architectures

Emerging deep learning architectures such as Vision Transformers (ViTs), hybrid CNN–Transformer models, and graph neural networks offer superior capability in capturing global and local image features. These models can better analyze subtle retinal changes and structural patterns associated with IRDs. Additionally, self-supervised and semi-supervised learning techniques will allow models to learn from large volumes of unlabeled medical data, addressing the challenge of limited annotated datasets.

3. Early-Stage Disease Detection

Future AI-based systems will focus on detecting inherited retinal diseases at very early or pre-symptomatic stages. By identifying minor microstructural changes in retinal layers, AI models can alert clinicians before noticeable vision impairment occurs. Early detection is crucial for monitoring disease progression and initiating timely interventions to preserve vision.

4. Disease Severity and Progression Prediction

Beyond diagnosis, AI can be used to predict disease severity and progression trends over time. Longitudinal analysis of retinal images will help estimate degeneration rates and future visual outcomes. Such predictive capabilities will support clinicians in designing personalized treatment plans, scheduling follow-ups, and managing patient expectations.

5. Real-Time and Edge AI Deployment

Advancements in lightweight models and hardware acceleration will enable AI deployment on edge devices such as portable fundus cameras and smartphones. Real-time analysis will facilitate mass screening programs, particularly in rural or underserved areas where ophthalmology specialists are scarce. This democratization of healthcare technology can significantly expand access to early retinal disease diagnosis.

6. Large-Scale Clinical Validation and Regulatory Acceptance

For widespread clinical adoption, future AI systems will undergo extensive multi-center validation across diverse populations. Large-scale clinical trials will help assess model robustness, minimize demographic bias, and ensure safety and reliability. These validation efforts will also support regulatory approval and integration into standard clinical workflows.

7. Personalized Treatment and Gene Therapy Support

AI systems can assist in identifying patients who are suitable candidates for gene therapy or targeted treatments by analyzing retinal features alongside genetic profiles. This capability will accelerate the shift toward precision ophthalmology, enabling individualized treatment strategies that maximize therapeutic effectiveness and minimize risks.

8. Continuous Learning and Adaptive Systems

Future AI platforms will incorporate continuous and federated learning mechanisms, allowing models to improve over time as new patient data and disease variants emerge. Such adaptive systems will ensure that diagnostic performance remains up-to-date with evolving clinical knowledge and population trends.

9. Enhanced Explainability and Ethical AI

Explainable AI (XAI) will play a critical role in increasing clinician trust and acceptance. Visualization tools such as heatmaps and attention maps will help ophthalmologists understand how AI models arrive at their decisions. Ethical AI frameworks will also ensure fairness, transparency, data privacy, and accountability in clinical applications

REFERENCES

[1] S. R. Flaxman, R. R. A. Bourne, S. Resnikoff, P. Ackland, T. Braithwaite, M. V. Cicinelli, A. Das, J. B. Jonas, J. Keeffe, J. H. Kempen, J. Leasher, H. Limburg, K. Naidoo, K. Pesudovs, A. Silvester, G. A. Stevens, N. Tahhan, T. Y. Wong, H. R. Taylor, Global causes of blindness and distance vision impairment 1990-2020: a systematic review and meta-analysis, Lancet Glob Health 5 (2017) e1221–e1234.doi:S2214-109X(17)30393-5[pii]10.1016/S2214-109X(17)30393-5[doi].

[2] J. Köberlein, K. Beifus, C. Schaffert, R. P. Finger, The economic burden of visual impairment and blindness: a systematic review, BMJ Open 3 (2013) e003471.

[3] D. L. Demmin, S. M. Silverstein, Visual impairment and mental health: unmet needs and treatment options, Clinical Ophthalmology (2020) 4229–4251.

[4] J. R. Ehrlich, J. Ramke, D. Macleod, H. Burn, C. N. Lee, J. H. Zhang, W. Waldock, B. K. Swenor, I. Gordon, N. Congdon, M. Burton, J. R. Evans, Association between vision impairment and mortality: a systematic review and meta-analysis, Lancet Glob Health 9 (2021) e418–e430.doi:S2214-109X(20)30549-0[pii]10.1016/S2214-109X(20)30549-0[doi].

[5] M. Hanany, C. Rivolta, D. Sharon, Worldwide carrier frequency and genetic prevalence of autosomal recessive inherited retinal diseases, Proceedings of the National Academy of Sciences 117 (2020) 2710–2716.

[6] R. C. H. Jeffery, S. A. Mukhtar, I. L. McAllister, W. H. Morgan, D. A. Mackey, F. K. Chen, Inherited retinal diseases are the most common cause of blindness in the working-age population in australia, Ophthalmic Genetics 42 (2021) 431–439.

[7] RetNet, "Retinal Information Network," [Online]. Available: https://retnet.org/. [Accessed: Oct. 9, 2024].

[8] N. Schneider, Y. Sundaresan, P. Gopalakrishnan, A. Beryozkin, M. Hanany, E. Y. Levanon, E. Banin, S. Ben-Aroya, D. Sharon, Inherited retinal diseases: Linking genes, disease-causing variants, and relevant therapeutic modalities, Progress in Retinal and Eye Research 89 (2022) 101029.doi:https://doi.org/10.1016/j.preteyeres.2021.101029. URL https://www.sciencedirect.com/science/article/pii/S1350946221000902

[9] C. N. Keilhauer, F. C. Delori, Near-infrared autofluorescence imaging of the fundus: visualization of ocular melanin, Investigative Ophthalmology & Visual Science 47 (2006) 3556–3564.

[10] R. T. Smith, N. L. Gomes, G. Barile, M. Busuioc, N. Lee, A. Laine, Lipofuscin and autofluorescence metrics in progressive stgd, Investigative Ophthalmology & Visual Science 50 (2009) 3907–3914.

[11] S. Ting, L. Pasquale, L. Peng, J. P. Campbell, A. Y. Lee, R. Raman, G. S. Tan, L. Schmetterer, P. A. Keane, D. S. W. Ting, "Artificial intelligence and deep learning in ophthalmology," *British Journal of Ophthalmology* 103 (2019) 167–175. https://bjo.bmj.com/content/103/2/167

[12] A. Gulshan, L. Peng, M. Coram, M. C. Stumpe, D. Wu, A. Narayanaswamy, S. Venugopalan, K. Widner, T. Madams, J. Cuadros, R. Kim, R. Raman, P. C. Nelson, J. L. Mega, D. R. Webster, "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA* 316 (2016) 2402–2410. https://jamanetwork.com/journals/jama/fullarticle/2588763

[13] N. Grassmann, P. Mengelkamp, C. Brandl, S. Harsch, M. E. Zimmermann, B. Linkohr, A. Peters, C. Heid, M. H. Weber, T. Ach, A. K. Maier, B. H. F. Weber, "A deep learning algorithm for prediction of age-related eye disease study severity scale for age-related macular degeneration from color fundus photography," *Ophthalmology* 125 (2018) 1410–1420. https://www.sciencedirect.com/science/article/pii/S0161642018303529

[14] K. He, X. Zhang, S. Ren, J. Sun, "Deep residual learning for image recognition," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2016) 770–778.https://ieeexplore.ieee.org/document/7780459

[15] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature* 542 (2017) 115–118. https://www.nature.com/articles/nature21056

PUBLISHED PAPER

IJRTI Research Jou... 6 days ago

to me

	WhatsApp +919898157873		editor@ijrti.org		IJRTI

International Journal for Research Trends and Innovation - (IJRTI)

International Peer Reviewed & Refereed Journals, Open Access Journal

ISSN: 2456-3315 | Impact factor: 8.14 | ESTD Year: 2016

Scholarly open access journals, Peer-reviewed, and Refereed Journals, Impact factor 8.14 (Calculate by google scholar and Semantic Scholar | AI-Powered Research Tool) , Multidisciplinary, Monthly, Indexing in all major database & Metadata, Citation Generator, Digital Object Identifier(DOI)

Dear Author, Congratulation!!!

Your manuscript with Registration/Paper ID: 210897 has been Accepted for publication in the International Journal for Research Trends and Innovation (IJRTI) | IJRTI.ORG | ISSN: 2456-3315 | International Peer Reviewed & Refereed Journals, Open Access Online and Print Journal.

IJRTI Impact Factor: 8.14

Check Your Paper Status: track.php

Your Paper Review Report :

Registration/Paper ID:	210897				
Title of the Paper:	ARITIFICIAL INTELLIGENCE APPROACHES DETECTING INHERITED RETINAL DISEASES				
Unique Contents:	88% (Out of 100)	Paper Accepted:	Accepted	Overall Assessment (Comments):	Reviewer Comment in Online TRMS System

PLAGIARISM REPORT

turnitin Page 2 of 12 - AI Writing Overview Submission ID trn:oid:::2618:132597217

11% detected as AI

The percentage indicates the combined amount of likely AI generated text as well as likely AI generated text that was also likely AI paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

11 AI generated only 11%
Likely AI generated text from a large language model.

0 AI generated text that was AI paraphrased 0%
Likely AI generated text that was likely revised using an AI paraphrase tool or word spinner.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

turnitin Page 2 of 12 - AI Writing Overview Submission ID trn:oid:::2618:132597217